

Post-Harvest Loss Mitigation and Smart Storage Management Using IOT and Azure Digital Twin

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Abstract. Post-harvest loss poses a significant challenge to global food security, particularly in developing nations where storage inefficiencies lead to substantial wastage. This study proposes an IOT-enabled digital twin solution designed to minimize post-harvest losses through Smart storage management. Utilizing a Raspberry Pi integrated with DHT11 and MQ sensors, the system collects real-time environmental data, including temperature, humidity, and gas concentrations. The data is transmitted via Azure IOT Hub to a digital twin platform on Microsoft Azure, enabling real-time monitoring, simulation, and predictive analytics. Microsoft Azure multiple services analyze the data to provide actionable insights, facilitating proactive interventions to mitigate spoilage risks. By integrating low-cost hardware and scalable cloud services, this solution demonstrates the potential for transformative impact in reducing post-harvest waste and enhancing agricultural sustainability.

Keywords: Digital Twin , Post-Harvest, Smart Storage Management, Microsoft Azure

1 Introduction

With the rapidly growing global population projected to reach 9.7 billion by 2050 (UN, Population Division, 2022), about 70% extra food production will be required to feed them [1]. The agricultural sector faces numerous and unparalleled challenges. These include land scarcity, climate change, water scarcity, and resource depletion. Limited arable land hinders expansion of food production, while climate change disrupts agricultural processes through extreme weather events and altered patterns. Increasing competition for water resources for irrigation further compounds the issue, alongside the overuse of resources like soil nutrients, leading to decreased productivity. Addressing these challenges is crucial to ensure a sustainable and food-secure future for a growing global population [2]. This has resulted in widespread food insecurity, prompting researchers worldwide to propose various strategies and techniques aimed at increasing crop yields and promoting sustainable practices to secure global food supply [3]. This surge in population has placed immense pressure on the agricultural sector to significantly increase food production to meet the growing demand. To address this challenge and ensure food security, innovative solutions are necessary to optimize agricultural practices. In addition to these challenges, studies show that

more than 40% of food losses occur at the post-harvest stage in developing countries. These losses arise due to poor storage conditions, lack of environmental monitoring, and inadequate intervention mechanisms. Traditional storage management systems lack real-time tracking and predictive capabilities, leading to unnecessary spoilage. The motivation behind this research is to leverage IoT and Digital Twin technology to create an intelligent monitoring system that minimizes post-harvest losses. By implementing real-time data collection, predictive analytics, and automated alerts, this system aims to improve storage conditions and prevent spoilage before it occurs. This approach not only enhances food security but also contributes to reducing resource wastage, ensuring a more resilient and efficient agricultural supply chain.

Neme et al. [4] found that in developed countries, greater than 40% losses of food occurs at trade and customer stages, while in the case of developing countries, greater than 40% losses of food occur at post-harvest stage and processing point. Newly harvested high moisture unpreserved yields may quickly deteriorate owing to microorganism's attack. Post Harvest Loss (PHL) encompasses both quantitative (reduction in weight) and qualitative (loss of nutritional value or marketability) deterioration. The study emphasizes that losses are particularly high in developing countries, ranging from 20-40% for staple crops, translating to billions of dollars annually. 50%–60% cereal grains can be lost during the storage stage due only to the lack of technical inefficiency. This not only impacts food security but also reduces farmer income and contributes to food waste [1].

Researchers have turned to advanced solutions like Artificial Intelligence (AI), the Internet of Things (IOT) to tackle food security issues head-on. AI enhances agricultural productivity and sustainability through data-driven insights, optimizing practices, predicting outcomes, and automating tasks for greater efficiency. AI can be used in agriculture to improve food quality and security, reduce human intervention and effort, create a more transparent supply chain management system, optimize output and resource conservation through accurate predictions and precise decision-making [5]. IOT transforms farming by providing real-time sensor data for efficient resource management, remote monitoring, and improved post-harvest storage through automation and timely alerts. IOT introduces use of robots and drones (UAVs) in various agricultural settings, highlighting their impact on improving productivity and minimizing environmental impact [6]. Ndjuluwa et al. [7] looked into IOT and found that IOT also involve the use of sensors for monitoring factors like soil health, weather conditions, and plant growth. It also explores the integration of machine learning with IOT for tasks like weed detection and removal.

Among these transformative technologies, digital twin technology stands out as a cutting-edge solution addressing specific challenges such as post-harvest losses. A digital twin is a virtual replica of a physical object or system that uses real-time data to simulate, analyze, and optimize its real-world counterpart. It's used in various industries for monitoring performance, predicting outcomes, and improving operations. Key components include data collection, modeling, simulation, analysis, and feedback. Digital twins, virtual representations of physical

assets, are emerging as a promising solution to address these challenges. By integrating real-time data with crop growth models, digital twins can optimize resource management, detect and prevent diseases and pests, and predict yields, ultimately leading to more efficient, sustainable, and resilient agricultural practices. The objective of this study is to minimize post-harvest loss by leveraging an IOT-enabled digital twin solution. The digital twin will integrate with a Raspberry Pi and multiple sensors to simulate and monitor storage conditions in real-time. Data collected from these sensors will then be analyzed to enable predictive insights for storage management and reduce post-harvest waste effectively. This present study implements digital twin for minimizing post-harvest losses.

2 Literature Review

Tagarakis et al. [8] investigated on implementing digital twin for irrigation system, where IoT and sensors were combined to collect data from field. Promising results were obtained in the study where the model obtained 90% accuracy in detecting the water shortage. Recent research highlights the transformative potential of digital twins (DTs) in agriculture. Tagarakis et al.[8] and Albahar et al. [9] explored how digital twins (DTs) act as virtual replicas of farming systems, enabling real-time monitoring, optimizing resource utilization, and enhancing crop yields. In their study, Ali Ikram et al. [10] delved into the application of DTs for precision farming, emphasizing their role in simulating scenarios and testing management strategies. Promising results were also highlighted by Albahar et al. [9], who discussed the integration of DTs with other digital platforms, fostering improved traceability and decision-making across the agricultural value chain. Finally, Verdouw et al. [11] emphasized the role of predictive analytics within DTs, allowing farmers to anticipate potential issues and implement preventative measures. These studies collectively showcased the diverse applications of DTs in revolutionizing agricultural practices.

Several studies have highlighted the use of advanced technologies in agriculture to improve efficiency and sustainability. Gonzalez et al.[12] focused on controlled environment agriculture by monitoring greenhouse operations using DHT sensors integrated with a Python program, leading to improved soil and crop monitoring with reduced human intervention. Verdouw et al. [13] explored smart farming water management by developing a digital twin based on soil probe data to optimize irrigation strategies, promoting water conservation and enhanced crop health. Another study emphasized the integration of digital twins and smart agricultural practices to improve crop cultivation, resource utilization, and management, achieving real-time monitoring and predictive analytics for sustainable farming. Research by Ding et al. [14] examined precision agriculture, specifically orchestrating digital twin systems to support decision-making and resource optimization, leveraging IoT data for enhanced crop health. Furthermore, precision nitrogen fertilization was addressed by integrating a crop growth model with real-time soil and weather data to determine optimal ni-

trogen application, reducing environmental impact and maximizing yield with minimal nitrogen waste.

3 Methodology

3.1 Components of the IOT system

Sensors and Controllers -

1. Raspberry Pi- The Raspberry Pi is a type of single-board computer (SBC). The board of a Raspberry Pi contains a CPU; memory; LAN, USB and micro HDMI ports; and a slot for a micro SD card. It acts as the central hub for collecting and processing sensor data. In this research, the Raspberry Pi 3 Model B is selected as the primary data processing and communication hub for the IOT-enabled digital twin solution because it is widely used in IOT nowadays.

Quad-core ARM Cortex-A53 CPU processor running at 1.2 GHz, to handle data from multiple sensors. 1 GB LPDDR2, which allows running applications like the operating system and data processing tasks. MicroSD card slot for storage, enabling you to store the OS and application files. 10/100 Ethernet port and built-in Wi-Fi (802.11n) and Bluetooth 4.2 support, enabling seamless data transmission. GPIO (General Purpose Input/Output) Pins for Sensor Connections. The Raspberry Pi 3 Model B comes equipped with 40 GPIO pins, which allow you to interface directly with various sensors (temperature, humidity, gas, etc.) using both digital and analog signals. These GPIO pins are arranged in a convenient, accessible layout, making it easy to connect a variety of external sensors and actuators.

2. Temperature Sensor (DHT11) - The DHT11 (Digital Humidity and Temperature) temperature sensor is used to monitor the ambient temperature within the storage environment. The DHT11 measures temperatures ranging from 0°C to 50°C with a precision of $\pm 2^\circ\text{C}$, which is suitable for standard storage environments. The sensor is connected to the Raspberry Pi using its GPIO pins. The DHT11 sensor is strategically placed within the storage unit to ensure it captures an accurate representation of the overall temperature.

3. Humidity Sensor (DHT11) - Alongside temperature, the DHT11 also measures humidity, which is critical in preventing spoilage caused by excessive moisture. The humidity data is gathered simultaneously with temperature readings. The DHT11 measures relative humidity from 20% to 90% with an accuracy of $\pm 5\%$. Although it has limitations in precision, it is adequate for monitoring general trends in moisture levels within the storage facility.

4. Gas Sensors (MQ Sensor) - The MQ series sensor, MQ-135 sensor used in this project, is designed to detect the presence of gases like ethylene. The sensor detects gas concentrations and sends analog signals to the Raspberry Pi, which are then converted into digital values using an analog-to-digital converter

(ADC). The MQ sensor can detect a range of gases, including ethylene, and provides real-time data on gas concentrations. The MQ sensor is connected to the Raspberry Pi through a series of wiring connections, and an ADC is used to interpret the analog signals, refer Fig. 1.

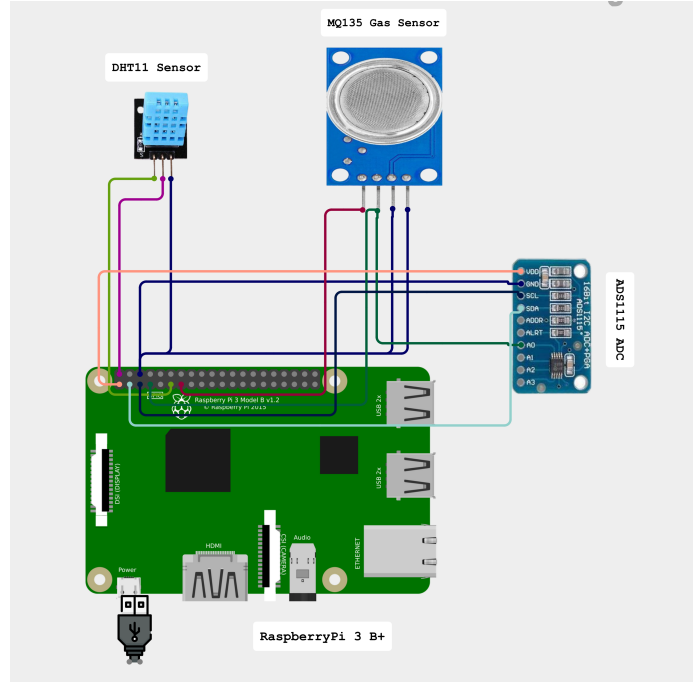


Fig. 1. Hardware Architecture

3.2 Digital Twin Platform

Similar to other digitalization concepts, such as the internet of things (IoT), cloud computing, machine learning/artificial intelligence, and augmented reality, the Digital Twin concept has seen increasing interest in recent years in both academia and industry as indicated by the growth of publications, articles, and commercial marketing [15].

To create an effective and scalable IOT-enabled digital twin solution for post-harvest storage management, architecture leverages Microsoft Azure [16] multiple service i.e. Azure Digital Twin service, Azure IOT Hub, Azure IOT Central App, Azure Blob Storage, Azure service bus and Logic app as shown in Fig.2. These Azure services play critical role in our study in enabling real-time monitoring, simulation and data analysis, contributing significantly to minimizing

post-harvest loss.

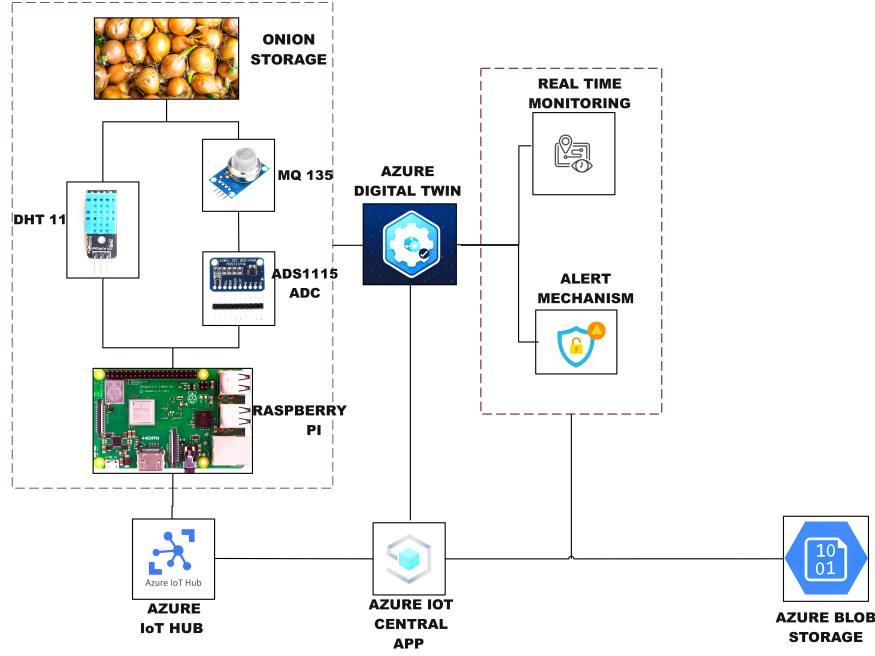


Fig. 2. Overview of our proposed end-to-end workflow of IOT system

Azure Digital Twin Service - Azure Digital Twin is a powerful platform for creating digital models of real-world environments. In our research, it is used to build a virtual representation of the physical storage environment integrated with data from Raspberry Pi sensors. By simulating the environmental conditions and storage scenarios, Azure Digital Twin enables real-time monitoring and predictive analytics to identify potential post-harvest issues, such as temperature or gas fluctuations, before they lead to loss.

Azure IOT Hub - Azure IOT Hub serves as the central gateway for secure, bi-directional communication between the Raspberry Pi (with attached DHT11 and MQ sensors) and the cloud. Sensor data, including temperature, humidity, and gas concentrations, is transmitted to IOT Hub for further analysis and storage. This seamless data flow ensures timely updates, enabling the predictive management of storage conditions.

Azure IOT Central App - Azure IOT Central App simplifies the management and visualization of IOT devices and data. In the present study, IOT

Central provides a user-friendly interface to monitor the health of connected Raspberry Pi devices and track sensor readings in real time. This platform allows stakeholders to easily view insights, such as environmental anomalies, and make informed decisions to prevent spoilage.

Azure Blob Storage - Azure Blob Storage is utilized for storing large volumes of sensor data and historical environmental records. The raw sensor data collected from IOT Hub is stored in Blob Storage, enabling cost-effective and secure data retention. This data is later processed for training machine learning models to predict optimal storage conditions and minimize post-harvest losses.

Azure Service Bus - Azure Service Bus acts as a messaging platform to ensure reliable communication between different Azure services. Here Service Bus helps orchestrate workflows, such as sending alerts when critical environmental thresholds are crossed. It ensures that messages are delivered seamlessly from the IOT Hub to downstream applications or decision-making systems.

Azure Logic App - Azure Logic App automates workflows by integrating different Azure services. Logic App triggers alerts or actions based on sensor data anomalies. For example, if temperature or gas readings exceed safe limits, Logic App automatically sends notifications via email or dashboards to stakeholders, allowing them to take corrective measures promptly.

4 Results and Discussion

4.1 Data gathering and preprocessing

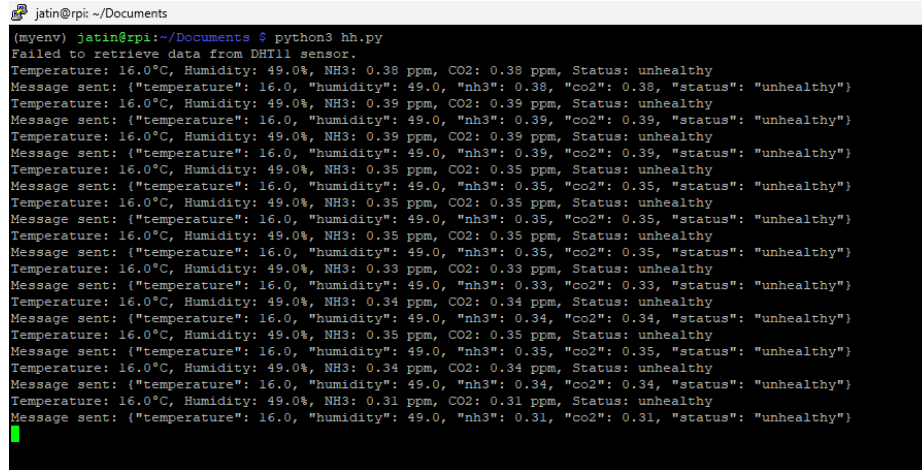
This study implements a real-time environmental monitoring system using a combination of hardware and software components, interfacing sensors with a Raspberry Pi and integrating the collected data into a cloud-based digital twin model.

Hardware Implementation: The system uses a DHT11 sensor to monitor temperature and humidity, and an MQ135 sensor for detecting air quality, specifically monitoring NH_3 (ammonia) and CO_2 (carbon dioxide) levels. These are critical indicators for determining spoilage or an unhealthy environment in post-harvest storage. Since the MQ135 sensor provides analog output, an Analog-to-Digital Converter (ADC) is used to convert the data into a digital format before sending it to the Raspberry Pi. The sensors are physically connected to the Raspberry Pi via General Purpose Input/Output (GPIO) pins. The DHT11 sensor directly communicates using a digital signal, whereas the MQ135 sensor requires an intermediate ADC to enable digital data acquisition.

Software Implementation & Data Processing: Python scripts running on the Raspberry Pi continuously poll sensor data at regular intervals. For this ,

The AdafruitDHT library is used for retrieving temperature and humidity readings from the DHT11 sensor. The MQ135 sensor readings are acquired using an ADC module and processed through Python-based analog-to-digital conversion methods. If a sensor fails to provide readings, the system logs the failure and attempts retries to ensure data consistency. Once the sensor data is collected, it is formatted into structured JSON objects and sent to Raspberry Pi as in Fig. 3 This ensures compatibility with further processing steps and seamless integration with other IoT components. The Raspberry Pi acts as a local hub for aggregation and preliminary processing of data before transmission to cloud services.

Integration with Cloud & Digital Twin: The environmental parameters are continuously assessed against predefined safety thresholds. If Temperature falls outside the range of 25-30°C [17], Humidity exceeds 65-70%, NH₃ or CO₂ concentrations surpass safe limits, the system flags the environment as “unhealthy.” This analysis helps predict spoilage risks or hazardous storage conditions, enabling early intervention. To ensure real-time synchronization between physical storage conditions and the digital twin, the processed data is transmitted from the Raspberry Pi to Azure IoT Hub as in Fig 4. The IoT Hub acts as a gateway, transferring the information to the Azure Digital Twin Explorer for visualization, simulation, and further predictive analytics. The data can also be displayed on a dashboard or trigger alerts when unsafe conditions are detected.



```
jatin@rpil: ~/Documents
(myenv) jatin@rpil:~/Documents $ python3 hh.py
Failed to retrieve data from DHT11 sensor.
Temperature: 16.0°C, Humidity: 49.0%, NH3: 0.38 ppm, CO2: 0.38 ppm, Status: unhealthy
Message sent: {"temperature": 16.0, "humidity": 49.0, "nh3": 0.38, "co2": 0.38, "status": "unhealthy"}
Temperature: 16.0°C, Humidity: 49.0%, NH3: 0.39 ppm, CO2: 0.39 ppm, Status: unhealthy
Message sent: {"temperature": 16.0, "humidity": 49.0, "nh3": 0.39, "co2": 0.39, "status": "unhealthy"}
Temperature: 16.0°C, Humidity: 49.0%, NH3: 0.39 ppm, CO2: 0.39 ppm, Status: unhealthy
Message sent: {"temperature": 16.0, "humidity": 49.0, "nh3": 0.39, "co2": 0.39, "status": "unhealthy"}
Temperature: 16.0°C, Humidity: 49.0%, NH3: 0.39 ppm, CO2: 0.39 ppm, Status: unhealthy
Message sent: {"temperature": 16.0, "humidity": 49.0, "nh3": 0.39, "co2": 0.39, "status": "unhealthy"}
Temperature: 16.0°C, Humidity: 49.0%, NH3: 0.35 ppm, CO2: 0.35 ppm, Status: unhealthy
Message sent: {"temperature": 16.0, "humidity": 49.0, "nh3": 0.35, "co2": 0.35, "status": "unhealthy"}
Temperature: 16.0°C, Humidity: 49.0%, NH3: 0.35 ppm, CO2: 0.35 ppm, Status: unhealthy
Message sent: {"temperature": 16.0, "humidity": 49.0, "nh3": 0.35, "co2": 0.35, "status": "unhealthy"}
Temperature: 16.0°C, Humidity: 49.0%, NH3: 0.35 ppm, CO2: 0.35 ppm, Status: unhealthy
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Message sent: {"temperature": 16.0, "humidity": 49.0, "nh3": 0.35, "co2": 0.35, "status": "unhealthy"}
Temperature: 16.0°C, Humidity: 49.0%, NH3: 0.33 ppm, CO2: 0.33 ppm, Status: unhealthy
Message sent: {"temperature": 16.0, "humidity": 49.0, "nh3": 0.33, "co2": 0.33, "status": "unhealthy"}
Temperature: 16.0°C, Humidity: 49.0%, NH3: 0.34 ppm, CO2: 0.34 ppm, Status: unhealthy
Message sent: {"temperature": 16.0, "humidity": 49.0, "nh3": 0.34, "co2": 0.34, "status": "unhealthy"}
Temperature: 16.0°C, Humidity: 49.0%, NH3: 0.35 ppm, CO2: 0.35 ppm, Status: unhealthy
Message sent: {"temperature": 16.0, "humidity": 49.0, "nh3": 0.35, "co2": 0.35, "status": "unhealthy"}
Temperature: 16.0°C, Humidity: 49.0%, NH3: 0.34 ppm, CO2: 0.34 ppm, Status: unhealthy
Message sent: {"temperature": 16.0, "humidity": 49.0, "nh3": 0.34, "co2": 0.34, "status": "unhealthy"}
Temperature: 16.0°C, Humidity: 49.0%, NH3: 0.31 ppm, CO2: 0.31 ppm, Status: unhealthy
Message sent: {"temperature": 16.0, "humidity": 49.0, "nh3": 0.31, "co2": 0.31, "status": "unhealthy"}

```

Fig. 3. Data Reading by Sensor network at regular intervals being sent to Raspberrypi

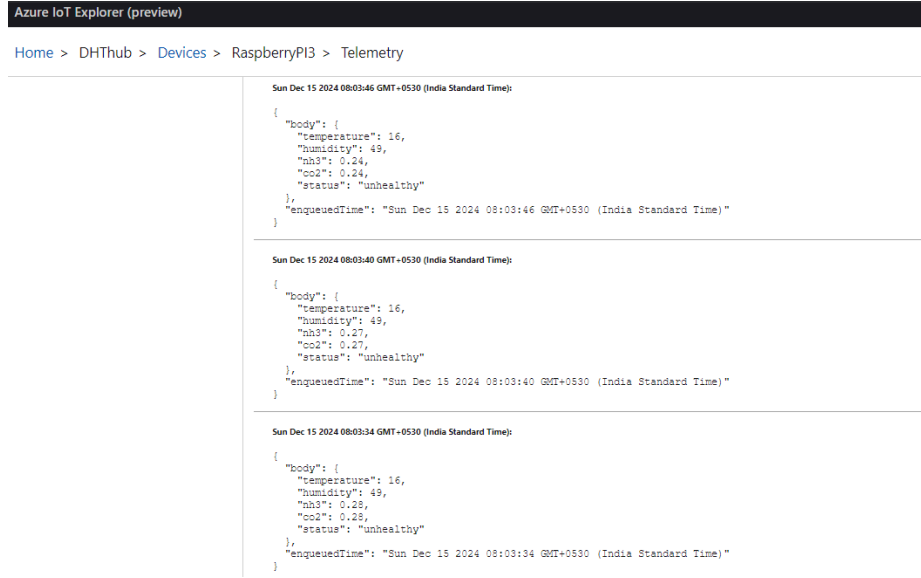


Fig. 4. Reading from raspberrypi are sent to Azure Explorer in JSON format

4.2 Real-Time Monitoring and Alerts-

The real-time monitoring and alert system ensures that storage conditions are continuously tracked and provides timely notifications when parameters deviate from safe thresholds. This system integrates sensor networks, cloud services, and automated notification mechanisms. The system consists of interconnected sensors (DHT11 and MQ135) that continuously collect environmental data, including: Temperature ($^{\circ}\text{C}$), Humidity (%), Gas Concentrations (NH_3 , CO_2). The sensor data is transmitted to the Raspberry Pi via GPIO pins, formatted into JSON objects, and securely sent to Azure IoT Hub using Python scripts. The Raspberry Pi ensures periodic updates to the cloud, maintaining real-time monitoring capabilities.

Cloud Processing & Threshold Analysis: The Azure IoT Hub serves as the cloud-based gateway for receiving, managing, and monitoring IoT data from the Raspberry Pi. It processes incoming sensor data and compares it against predefined threshold values: if Temperature $> 30^{\circ}\text{C} \rightarrow$ Trigger Alert, if $\text{NH}_3 > 0.3 \text{ ppm} \rightarrow$ Trigger Alert. If any threshold is breached, the IoT Hub routes the data to the Azure Logic App for automated alert generation.

Automated Alert Mechanism: The Azure Logic App listens for messages from the IoT Hub and evaluates sensor data for unsafe conditions. When a threshold is breached, the Logic App triggers an automated workflow that sends an email notification to concerned stakeholders. The email contains: Current

Sensor Readings , Alert Status (“Unhealthy” or “Threshold Breached”) , Immediate actions required to prevent post-harvest loss. This mechanism enables real-time tracking of storage conditions and quick response actions, significantly minimizing post-harvest losses.

4.3 Alert Mechanism

The alert system is designed to promptly notify users when storage conditions become critical. Alerts are generated based on predefined criteria for each monitored parameter.

Temperature & Humidity Alerts: If temperature or humidity levels rise above or drop below safe storage conditions, an alert is generated. This prevents spoilage or dehydration of stored produce.

Gas Concentration Alerts: If the MQ135 sensor detects high levels of ammonia, carbon dioxide, or ethylene—indicating accelerated ripening or spoilage ,an immediate alert is triggered. These alerts are visualized on the Azure Digital Twin platform and transmitted to users via automated notifications, ensuring proactive intervention to maintain optimal storage conditions. By integrating hardware (Raspberry Pi, sensors, ADC) with cloud-based software (Azure IoT Hub, Digital Twin, Logic Apps), this system enables a seamless, real-time monitoring framework for post-harvest storage management.

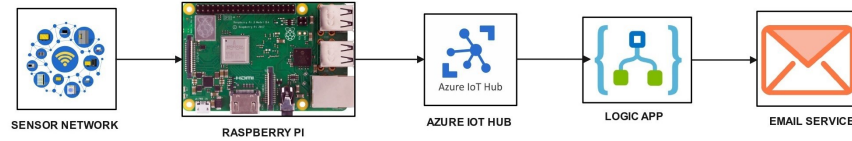


Fig. 5. Alert Mechanism Workflow

Notification System: Alerts are delivered to users through various channels to ensure they are informed promptly and can take necessary actions. These channels may include SMS Notifications where Text messages are sent to registered phone numbers. It can be Email Alerts where Detailed email notifications are sent with information about the detected issue, including data summaries and suggested corrective actions. If a mobile or web app is part of the system, push notifications are used to deliver alerts directly to users’ devices via app or web page.

For real time monitoring, the smartphone is integrated with the IOT System Using IOT Central , we registered a mobile phone as a device within Microsoft Azure IOT Central. This integration demonstrates the system’s ability to communicate with a variety of devices, facilitating real-time telemetry data collection and visualization. The mobile phone, serving as an IOT device, streams telemetry data such as temperature , humidity ,gas concentration levels , location,

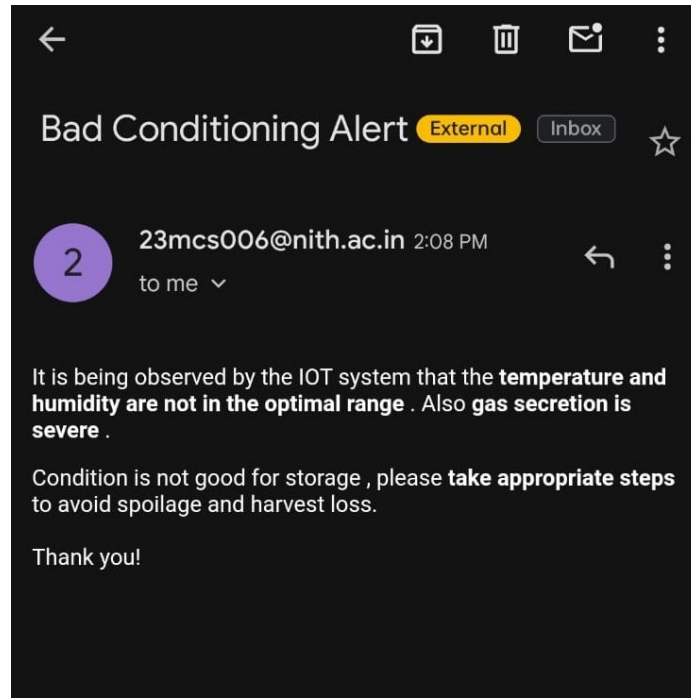


Fig. 6. Alert Notification from Email via Logic app

rotation and battery level to the IOT Central application. The mobile phone is provisioned in IOT Central under some name. Telemetry data is successfully transmitted, as evidenced by the raw data logs in the system. Telemetry data, including event timestamps and message types, is continuously streamed from the mobile phone to the IOT Central dashboard and vice-versa. The data is utilized to monitor environmental parameters or device-specific metrics. Fig. 7 shows the Azure IoT Central dashboard for a connected device named "my-phone-as-device-ieslpei-KM," displaying raw telemetry data, including timestamps and message types, with fields for metrics like acceleration and battery level.

Dashboard Visualization: Collected data is visualized using pre-configured templates within IOT Central for real-time analysis. Dashboard involves sensor widgets which display real-time data streams from connected sensors, including DHT11 Temp/Humidity Sensor and MQ135 Gas Sensor , Acceleration Sensor which Monitors movement or orientation changes. Dashboard also have a location Widget that tracks the geographic position of the mobile device, useful for spatial analysis in IOT applications like logistics or asset tracking.

This integration highlights the practical application of mobile phones in IOT systems as cost-effective and scalable devices for data collection. The mobile device's successful registration and real-time telemetry data transmission validate the system's interoperability and potential to monitor environmental conditions.

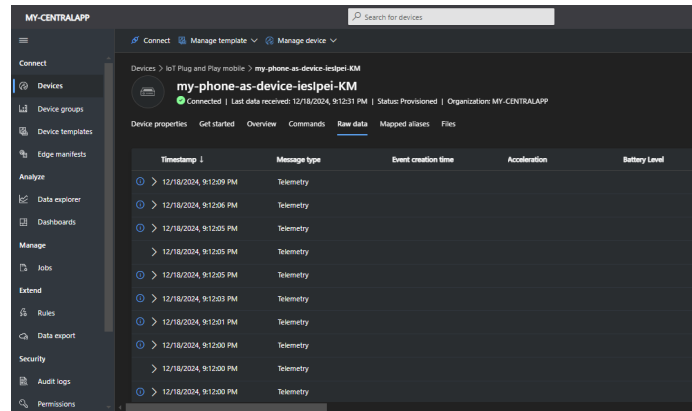


Fig. 7. Mobile Phone registered with IOT system using IOT Plug and Play.

Furthermore, these insights directly contribute to the research on IOT-enabled digital twin solutions for post-harvest loss minimization, where real-time data is critical for decision-making. By registering the mobile phone as an IOT device in Microsoft Azure IOT Central, it can act as a sensor node that collects, transmits, and processes telemetry data in real-time. The telemetry data is visualized and analyzed through Azure IOT Central dashboards or forwarded to other Azure services like Digital Twin for modeling. The mobile device integrates seamlessly with other IOT nodes, such as Raspberry Pi and additional sensors (DHT11, MQ135) and provides lot of advantages like -

Low-Cost, Scalable Sensor Node - Using a mobile phone as an IOT device eliminates the need for additional hardware for certain functions (e.g., location tracking, movement monitoring, or basic environmental sensing). This cost-efficient approach enables you to monitor transportation and storage condi-

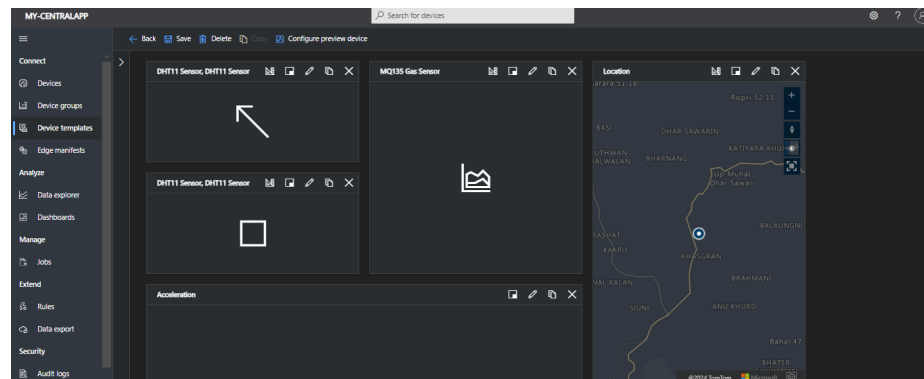


Fig. 8. Dashbaord of IOT central app connected with Mobile Phone

tions of agricultural produce and tracking the location and movement of storage bins or vehicles in real time.

Backup and Redundancy - The mobile phone serves as a reliable backup or secondary sensor node in your IOT ecosystem. For example: If a primary IOT device like the Raspberry Pi fails, the mobile device can ensure continuity in data collection, preventing data gaps that could hinder predictive insights.

Enable Farmer Participation - Farmers can use their mobile phones as temporary IOT nodes to monitor conditions if they lack access to advanced IOT infrastructure.

5 Conclusion

This research demonstrates the potential of an IOT-enabled digital twin solution to significantly reduce post-harvest losses by leveraging real-time environmental monitoring and smart storage management. The integration of Raspberry Pi with DHT11 and MQ sensors, coupled with Azure Digital Twin services, provides a scalable and cost-effective system for addressing the critical challenges in agricultural storage. By offering actionable insights through real-time data visualization and automated alerts, the system empowers stakeholders to take proactive measures to minimize spoilage.

In the future, we aim to incorporate advanced machine learning techniques into the system to enhance predictive analytics capabilities. By analyzing historical and real-time data, these models will enable more precise forecasting of spoilage risks, optimizing storage strategies, and further reducing post-harvest losses. Expanding the sensor network for more comprehensive monitoring, and scaling the system for deployment in larger storage facilities to ensure its effectiveness in real-world conditions. These integration will contribute to a smarter and more adaptive agricultural system, promoting sustainability and strengthening food security.

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